

60.6.5 Cortex-M7 embedded trace macrocell (ETM)

The Cortex-M7 ETM is a CoreSight™ component closely coupled to the CPU. The ETM generates trace packets that allow the execution of the Cortex-M7 core to be traced. In the STM32H7, the ETM is configured for instruction trace only, so data accesses are not included in the trace information.

The ETM receives information from the CPU over the processor trace interface, including:

- The number of instructions executed in the same cycle
- Changes in program flow
- The current processor instruction state
- The addresses of memory locations accessed by load and store instructions
- The type, direction and size of a transfer
- Condition code information
- Exception information
- Wait for interrupt state information

For more information, refer to the Arm® CoreSight™ ETM™-M7 technical reference manual [\[6\]](#).

Cortex-M7 ETM registers

ETM programming control register (M7_ETM_PRGCTL)

Address offset: 0x004

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EN
															rw

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **EN**: Trace program enable

0: Trace unit is disabled

1: Trace unit is enabled

ETM processor select control register (M7_ETM_PROCSEL)

Address offset: 0x008

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PROC SEL
															rw

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **PROCSEL**: Processor select

This field has no effect since only the Cortex-M7 uses this ETM.

ETM status register (M7_ETM_STAT)

Address offset: 0x00C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PMSTA BLE	IDLE
														r	r

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **PMSTABLE**: Programmers model stable

Indicates whether the ETM registers are stable and can be read.

0: Registers are not stable

1: Registers are stable

Bit 0 **IDLE**: Trace unit inactive

0: ETM is not idle

1: ETM is idle

ETM trace configuration register (M7_ETM_CONFIG)

Address offset: 0x010

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DV	DA
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	RS	TS	COND[2:0]			Res.	Res.	Res.	CCI	BB	INSTP0[1:0]		Res.
			rw	rw	rw	rw	rw				rw	rw	r	r	

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **DV**: Data value tracing (read-only)

0: Disabled

Bit 16 **DA**: Data address tracing (read-only)

0: Disabled

Bits 15:13 Reserved, must be kept at reset value.

Bit 12 **RS**: Return stack enable

0: Disabled

1: Enabled

Bit 11 **TS**: Global timestamp tracing

0: Disabled

1: Enabled

Bits 10:8 **COND[2:0]**: Conditional instruction tracing

0x0: Conditional instruction tracing disabled

0x1: Conditional load instructions are traced

0x2: Conditional store instructions are traced

0x3: Conditional load and store instructions are traced

0x7: All conditional instructions are traced

Other: Reserved

Bits 7:5 Reserved, must be kept at reset value.

Bit 4 **CCI**: Cycle counting in instruction trace

0: Disabled

1: Enabled

Bit 3 **BB**: Branch broadcast mode

0: Disabled

1: Enabled

Bits 2:1 **INSTP0[1:0]**: Determines which instructions are P0 instructions (read-only)

0x0: Only branches are P0 instructions

Bit 0 Reserved, must be kept at reset value.

ETM event control 0 register (M7_ETM_EVENTCTL0)

Only accepts writes when trace unit is disabled

Address offset: 0x020

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TYPE1	Res.	Res.	Res.	SEL1[3:0]				TYPE0	Res.	Res.	Res.	SEL0[3:0]			
rw				rw	rw	rw	rw	rw				rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bit 15 **TYPE1**: Resource type for event 1

0: Single selected resource

1: Boolean combined resource pair

Bits 14:12 Reserved, must be kept at reset value.

Bits 11:8 **SEL1[3:0]**: Resource / Boolean combined resource pair, for event 1

When TYPE1 is 0, selects a single selected resource from 0-15 defined by bits[3:0]

When TYPE1 is 1, selects a Boolean combined resource pair from 0-7 defined by bits[2:0]

Bit 7 **TYPE0**: Resource type for event 0

0: Single selected resource

1: Boolean combined resource pair

Bits 6:4 Reserved, must be kept at reset value.

Bits 3:0 **SEL0[3:0]**: Resource / Boolean combined resource pair for event 0

When TYPE0 is 0, selects a single selected resource from 0-15 defined by bits[3:0]

When TYPE0 is 1, selects a Boolean combined resource pair from 0-7 defined by bits[2:0]

ETM event control 1 register (M7_ETM_EVENTCTL1)

Only accepts writes when trace unit is disabled

Address offset: 0x024

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	LPOVE RRIDE	ATB	Res.	Res.	Res.	Res.	Res.	Res.	Res.	INSTEN[3:0]			
			rw	rw								rw	rw	rw	rw

Bits 31:13 Reserved, must be kept at reset value.

Bit 12 **LPOVERRIDE**: Low power state behavior override

0: Low power state normal behavior

1: Entry to low power state does not affect resources and event trace generation

Bit 11 **ATB**: ATB trigger enable

0: Disabled

1: Enabled

Bits 10:4 Reserved, must be kept at reset value.

Bits 3:0 **INSTEN[3:0]**: Instruction trace event element enable

Each bit corresponds to an event:

0bXXX0: Event 0 does not cause an event element

0bXXX1: Event 0 causes an event element

0bXX0X: Event 1 does not cause an event element

0bXX1X: Event 1 causes an event element

0bX0XX: Event 2 does not cause an event element

0bX1XX: Event 2 causes an event element

0b0XXX: Event 3 does not cause an event element

0b1XXX: Event 3 causes an event element

ETM stall control register (M7_ETM_STALLCTL)

Only accepts writes when trace unit is disabled

Address offset: 0x02C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	DSTALL	ISTALL	Res.	Res.	Res.	Res.	LEVEL[1:0]		Res.	Res.
						rw	rw					rw	rw		

Bits 31:10 Reserved, must be kept at reset value.

Bit 9 **DSTALL**: Stall processor based on data trace buffer space

0: Do not stall processor

1: Stall processor

Bit 8 **ISTALL**: Stall processor based on instruction trace buffer space

0: Do not stall processor

1: Stall processor

Bits 7:4 Reserved, must be kept at reset value.

Bits 3:2 **LEVEL[1:0]**: Stalling threshold level

A low level minimizes the amount of processor stalling, with a higher risk of FIFO overflow. A high level minimizes the risk of FIFO overflow but increases the amount of processor stalling.

Bits 1:0 Reserved, must be kept at reset value.

ETM global timestamp control register (M7_ETM_TSCTL)

Only accepts writes when trace unit is disabled

Address offset: 0x030

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TYPE	Res.	Res.	Res.	SEL[3:0]			
								rw				rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **TYPE**: Resource type for time stamp insertion

0: Single selected resource

1: Boolean combined resource pair

Bits 6:4 Reserved, must be kept at reset value.

Bits 3:0 **SEL[3:0]**: Resource / Boolean combined resource pair

When TYPE is 0, selects a single selected resource from 0-15 defined by bits[3:0]

When TYPE is 1, selects a Boolean combined resource pair from 0-7 defined by bits[2:0]

ETM synchronization period register (M7_ETM_SYNCP)

Address offset: 0x034

Reset value: 0x0000 000A

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PERIOD[4:0]				
											r	r	r	r	r

Bits 31:5 Reserved, must be kept at reset value.

Bits 4:0 **PERIOD[4:0]**: Trace bytes between synchronization requests

Defines the number of bytes of trace information between trace synchronization requests.

0xA: 1024 bytes

ETM cycle count control register (M7_ETM_CCCTL)

Address offset: 0x038

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	THRESHOLD[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **THRESHOLD[11:0]**: Threshold value for instruction trace cycle counting

The threshold represents the minimum interval between cycle count trace packets.

0x0: Reserved

Other: Threshold

ETM trace ID register (M7_ETM_TRACEID)

Address offset: 0x040

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TRACEID[6:0]						
									rw	rw	rw	rw	rw	rw	rw

Bits 31:7 Reserved, must be kept at reset value.

Bits 6:0 **TRACEID[6:0]**: Trace ID

0x00: Reserved

0x01 to 0x6F: Valid ID

0x70 to 0x7F: Reserved

ETM ViewInst main control register (M7_ETM_VICTL)

Address offset: 0x080

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EXLEV EL_S3	Res.	Res.	EXLEV EL_S0
												rw			rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TRCE RR	TRCR ESET	SSSTA TUS	Res.	TYPE	Res.	Res.	Res.	SEL[3:0]			
				rw	rw	rw		rw				rw	rw	rw	rw

Bits 31:20 Reserved, must be kept at reset value.

Bit 19 **EXLEVEL_S3**: Trace disable, exception level 3

Disables tracing in the specified exception level in Secure state for exception level 3.

0: Enable ViewInst in this exception level

1: Disable ViewInst in this exception level

Bits 18:17 Reserved, must be kept at reset value.

Bit 16 **EXLEVEL_S0**: Trace disable, exception level 0

Disables tracing in the specified exception level in Secure state for exception level 0.

0: Enable ViewInst in this exception level

1: Disable ViewInst in this exception level

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **TRCERR**: Tracing of system error exception

Selects whether a system error exception must always be traced.

0: System error exception is traced only if the instruction or exception immediately before the system error exception is traced

1: System error exception is always traced regardless of the value of ViewInst

Bit 10 **TRCRESET**: Tracing of reset exception

Selects whether a reset exception must always be traced.

0: Reset exception is traced only if the instruction or exception immediately before the reset exception is traced

1: Reset exception is always traced regardless of the value of ViewInst

Bit 9 **SSSTATUS**: Current status of the start/stop logic

0: Stop state

1: Started state

Bit 8 Reserved, must be kept at reset value.

Bit 7 **TYPE**: Resource type

0: Single selected resource

1: Boolean combined resource pair

Bits 6:4 Reserved, must be kept at reset value.

Bits 3:0 **SEL[3:0]**: Resource / Boolean combined resource pair

When TYPE is 0, selects a single selected resource from 0-15 defined by bits[3:0]

When TYPE is 1, selects a Boolean combined resource pair from 0-7 defined by bits[2:0]

ETM ViewInst start/stop control register (M7_ETM_VISSCTL)

Address offset: 0x088

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	STOP[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	START[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:16 **STOP[7:0]**: Selector of single address comparators to stop trace

Defines the single address comparators to stop trace with the ViewInst Start/Stop control.

One bit is provided for each implemented single address comparator.

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **START[7:0]**: Selector of single address comparators to start trace

Defines the single address comparators to start trace with the ViewInst Start/Stop control.

One bit is provided for each implemented single address comparator.

ETM ViewInst start/stop processor comparator control register (M7_ETM_VIPCSSCTL)

Address offset: 0x08C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	STOP[3:0]			
												rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	START[3:0]			
												rw	rw	rw	rw

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:16 **STOP[3:0]**: Selector of processor comparator input to stop trace

Selects which processor comparator inputs are in use with ViewInst start-stop control, for the purpose of stopping trace. One bit is provided for each processor comparator input.

Bits 15:4 Reserved, must be kept at reset value.

Bits 3:0 **START[3:0]**: Selector of processor comparator input to start trace

Selects which processor comparator inputs are in use with ViewInst start-stop control, for the purpose of starting trace. One bit is provided for each processor comparator input.

ETM counter reload value register (M7_ETM_CNTRLDV)

Address offset: 0x140

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALUE[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **VALUE[15:0]**: Counter reload value

This value is loaded into the counter each time the reload event occurs.

ETM ID register 8 (M7_ETM_IDR8)

Address offset: 0x180

Reset value: 0x0000 0002

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MAXSPEC[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MAXSPEC[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **MAXSPEC[31:0]**: Maximum speculation depth

Indicates the maximum speculation depth of the instruction trace stream. This is the maximum number of P0 elements that have not been committed in the trace stream at any one time.

0x2: Maximum trace speculation depth is 2

ETM ID register 9 (M7_ETM_IDR9)

Address offset: 0x184

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NUMP0KEY[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NUMP0KEY[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **NUMP0KEY[31:0]**: Number of P0 right-hand keys used
 0x0: No P0 keys used in instruction trace only configuration

ETM ID register 10 (M7_ETM_IDR10)

Address offset: 0x188

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NUMP1KEY[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NUMP1KEY[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **NUMP1KEY[31:0]**: Total number of P1 right-hand keys
 Indicates the total number of P1 right-hand keys, including normal and special keys.
 0x0: No P1 keys used in instruction trace only configuration

ETM ID register 11 (M7_ETM_IDR11)

Address offset: 0x18C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NUMP1SPC[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NUMP1SPC[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **NUMP1SPC[31:0]**: Total number of special P1 right-hand keys used
 0x0: No special P1 keys used

ETM ID register 12 (M7_ETM_IDR12)

Address offset: 0x190

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NUMCONDKEY[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NUMCONDKEY[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **NUMCONDKEY[31:0]**:

Indicates the total number of conditional instruction right-hand keys, including normal and special keys.

0x1: One conditional instruction right hand-key implemented

ETM ID register 13 (M7_ETM_IDR13)

Address offset: 0x194

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NUMCONDSPC[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NUMCONDSPC[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **NUMCONDSPC[31:0]**: Number of special conditional instruction right-hand keys

0x0: No special conditional instruction right hand-keys implemented

ETM implementation specific register 0 (M7_ETM_IMSPEC0)

Address offset: 0x1C0

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SUPPORT[3:0]			
												r	r	r	r

Bits 31:4 Reserved, must be kept at reset value.

Bits 3:0 **SUPPORT[3:0]**: Support for implementation specific extensions
0x0: No implementation specific extensions are supported

ETM ID register 0 (M7_ETM_IDR0)

Address offset: 0x1E0

Reset value: 0x0C00 1EE1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	COMM OPT	TSSIZE[4:0]					Res.	Res.	Res.	Res.	Res.	Res.	Res.	QSUPP [1]
		r	r	r	r	r	r								r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QSUP P[0]	Res.	CONDTYPE[1:0]]	NUMEVENT[1:0]]		RETST ACK	Res.	TRCCC I	TRCCO ND	TRCBB	Res.	Res.	Res.	Res.	Res.	Res.
r		r	r	r	r	r		r	r	r					

Bits 31:30 Reserved, must be kept at reset value.

Bit 29 **COMMOPT**: Meaning of the commit field in some packets
0: Commit mode 0

Bits 28:24 **TSSIZE[4:0]**: Global timestamp size
0x08: Maximum of 64-bit global timestamp implemented

Bits 23:17 Reserved, must be kept at reset value.

Bits 16:15 **QSUPP[1:0]**: Q element support
0x0: Q elements not supported

Bit 14 Reserved, must be kept at reset value.

Bits 13:12 **CONDTYPE[1:0]**: Conditional result tracing type
0x1: APSR condition flag values traced

Bits 11:10 **NUMEVENT[1:0]**: Number of events supported in the trace
0x1: Two events supported for instruction only configuration

Bit 9 **RETSTACK**: Return stack support
1: Two entry return stack supported

Bit 8 Reserved, must be kept at reset value.

Bit 7 **TRCCCI**: Support for cycle counting in the instruction trace
1: Cycle counting in the instruction trace is implemented

Bit 6 **TRCCOND**: Support for conditional instruction tracing
1: Conditional instruction trace is implemented

Bit 5 **TRCBB**: Support for branch broadcast tracing
1: Branch broadcast trace is implemented

Bits 4:0 Reserved, must be kept at reset value.

ETM ID register 1 (M7_ETM_IDR1)

Address offset: 0x1E4

Reset value: 0x4100 F401

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DESIGNER[7:0]								Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
r	r	r	r	r	r	r	r								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	TRCARCHMAJ[3:0]				TRCARCHMIN[3:0]				REVISION[3:0]			
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:24 **DESIGNER[7:0]**: Trace unit designer entity

0x41: Arm®

Bits 23:12 Reserved, must be kept at reset value.

Bits 11:8 **TRCARCHMAJ[3:0]**: Major trace unit architecture version number

0x4: ETM v4

Bits 7:4 **TRCARCHMIN[3:0]**: Minor trace unit architecture version number

0x0: Minor version 0

Bits 3:0 **REVISION[3:0]**: Implementation revision number

0x1: Rev 1

ETM ID register 2 (M7_ETM_IDR2)

Address offset: 0x1E8

Reset value: 0x0000 0004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	CCSIZE[3:0]				DVSIZE[4:0]				DASIZE[4:1]				
			r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DASIZE[0]	VMIDSIZE[4:0]					CIDSIZE[4:0]					IASIZE[4:0]				
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:29 Reserved, must be kept at reset value.

Bits 28:25 **CCSIZE[3:0]**: Cycle counter size

Indicates the size of the cycle counter in bits minus 12.

0x0: Cycle counter is 12 bits

Bits 24:20 **DVSIZE[4:0]**: Data value size in bytes

0x0: Data value size is not supported in instruction only configuration

Bits 19:15 **DASIZE[4:0]**: Data address size in bytes

0x0: Data address size is not supported in instruction only configuration

Bits 14:10 **VMIDSIZE[4:0]**: Virtual machine ID size
0x0: Virtual machine ID tracing not implemented

Bits 9:5 **CIDSIZE[4:0]**: Context ID size
0x0: Context ID tracing not implemented

Bits 4:0 **IASIZE[4:0]**: Instruction address size
0x4: 32-bit maximum address size

ETM ID register 3 (M7_ETM_IDR3)

Address offset: 0x1EC

Reset value: 0x0509 0004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NOOVERFLOW	NUMPROC[2:0]			SYSSTALL	STALLCTL	SYNCPR	TRCERR	Res.	Res.	Res.	Res.	EXLEVEL_S[3:0]			
r	r	r	r	r	r	r	r					r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	CCITMIN[11:0]											
				r	r	r	r	r	r	r	r	r	r	r	r

Bit 31 **NOOVERFLOW**: Support of NOOVERFLOW
Indicates whether the NOOVERFLOW of trace stall control is implemented.
0: Not implemented

Bits 30:28 **NUMPROC[2:0]**: Number of processors available for tracing
0x0: Only one processor can be traced

Bit 27 **SYSSTALL**: System support for stall control of the processor
0: Not supported

Bit 26 **STALLCTL**: Stall control support
1: Trace stall control (TRCSTALLCTL) is implemented

Bit 25 **SYNCPR**: Trace synchronization period support
0: TRCSYNCPR is read-only for instruction trace only configuration; the trace synchronization period is fixed

Bit 24 **TRCERR**: Support of TRCVICTLR.TRCERR
Indicates whether TRCVICTLR.TRCERR is implemented.
0x4: 32-bit maximum address size

Bits 23:20 Reserved, must be kept at reset value.

Bits 19:16 **EXLEVEL_S[3:0]**: Support of privilege levels
Privilege levels are implemented; one bit for each level.
0x9: Privilege levels Thread and Handler are implemented

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **CCITMIN[11:0]**: Instruction trace cycle counting minimum threshold
0x4: Minimum threshold is 4 instruction trace cycle

ETM ID register 4 (M7_ETM_IDR4)

Address offset: 0x1F0

Reset value: 0x0001 4000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NUMVMIDC[3:0]				NUMCIDC[3:0]				NUMSSCC[3:0]				NUMRSPAIR[3:0]			
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NUMPC[3:0]				Res.	Res.	Res.	SUPPAD DC	NUMDVC[3:0]				NUMACPAIRS[3:0]			
r	r	r	r				r	r	r	r	r	r	r	r	r

Bits 31:28 **NUMVMIDC[3:0]**: Number of Virtual Machine ID comparators implemented
0x0: None

Bits 27:24 **NUMCIDC[3:0]**: Number of Context ID comparators implemented
0x0: None

Bits 23:20 **NUMSSCC[3:0]**: Number of single-shot comparator controls implemented
0x0: None

Bits 19:16 **NUMRSPAIR[3:0]**: Number of resource selection pairs implemented
0x1: None

Bits 15:12 **NUMPC[3:0]**: Number of processor comparator inputs implemented
0x4: Four

Bits 11:9 Reserved, must be kept at reset value.

Bit 8 **SUPPADDC**: Support of data address comparisons
0: Not implemented

Bits 7:4 **NUMDVC[3:0]**: Number of data value comparators implemented
0x0: None

Bits 3:0 **NUMACPAIRS[3:0]**: Number of address comparator pairs implemented.
0x0: None

ETM ID register 5 (M7_ETM_IDR5)

Address offset: 0x1F4

Reset value: 0x90C7 0402

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REDF UNCN TR	NUMCNTR[2:0]			NUMSEQSTATE[2:0]			Res.	LPOVE RRIDE	ATBTRI G	TRACEIDSIZE[5:0]					
r	r	r	r	r	r	r		r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	NUMEXTINSEL[2:0]			NUMEXTIN[8:0]								
				r	r	r	r	r	r	r	r	r	r	r	r

- Bit 31 **REDFUNCNTR**: Support of reduced function counter
1: Implemented
- Bits 30:28 **NUMCNTR[2:0]**: Number of counters implemented
0x1: One counter implemented
- Bits 27:25 **NUMSEQSTATE[2:0]**: Number of sequencer states implemented
0x0: None
- Bit 24 Reserved, must be kept at reset value.
- Bit 23 **LPOVERRIDE**: Support of low-power state override
1: Implemented
- Bit 22 **ATBTRIG**: Support of ATB trigger
1: Implemented
- Bits 21:16 **TRACEIDSIZE[5:0]**: Number of trace ID bits
0x07: Seven-bit trace ID implemented.
- Bits 15:12 Reserved, must be kept at reset value.
- Bits 11:9 **NUMEXTINSEL[2:0]**: Number of implemented external input selectors
0x2: Two external input selectors implemented
- Bits 8:0 **NUMEXTIN[8:0]**: Number of implemented external inputs
0x2: Two external inputs implemented

ETM resource selection register 2 (M7_ETM_RSCTL2)

Address offset: 0x208

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PAIRIN V	INV	Res.	GROUP[2:0]		
										rw	rw		rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SELECT[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

- Bits 31:22 Reserved, must be kept at reset value.
- Bit 21 **PAIRINV**: Inversion of result of a combined pair of resources
0: Not inverted
1: Inverted
- Bit 20 **INV**: Inversion of the selected resources
0: Not inverted
1: Inverted
- Bit 19 Reserved, must be kept at reset value.

Bits 18:16 **GROUP[2:0]**: Selects a group of resources

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **SELECT[7:0]**: Selector of resources from desired group

Selects one or more resources from the desired group. One bit is provided per resource from the group.

ETM resource selection register 3 (M7_ETM_RSCTL3)

Address offset: 0x20C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	INV	Res.	GROUP[2:0]		
											rw		rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SELECT[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:21 Reserved, must be kept at reset value.

Bit 20 **INV**: Inversion of the selected resources

0: Not inverted

1: Inverted

Bit 19 Reserved, must be kept at reset value.

Bits 18:16 **GROUP[2:0]**: Selects a group of resources

Bits 15:8 Reserved, must be kept at reset value.

Bits 7:0 **SELECT[7:0]**: Selector of resources from desired group

Selects one or more resources from the desired group. One bit is provided per resource from the group.

ETM single-shot comparator control register 0 (M7_ETM_SSCC0)

Address offset: 0x280

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	RST	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
							rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:25 Reserved, must be kept at reset value.

Bit 24 **RST**: Single-shot comparator resource reset enable

Enables the single-shot comparator resource to be reset when it occurs, then enables another comparator match to be detected.

0: Disabled

1: Reset enabled; multiple matches can occur

Bits 23:0 Reserved, must be kept at reset value.

ETM single-shot comparator status register 0 (M7_ETM_SSCS0)

Address offset: 0x2A0

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STATUS	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DV	DA	INST
													r	r	r

Bit 31 **STATUS**: Single-shot status

This indicates whether any of the selected comparators have matched. If SSCC0.RST is set to 0, the STATUS bit must be written with 0 in order to enable single-shot comparator control.

0: No match occurred

1: Match has occurred at least once.

Bits 30:3 Reserved, must be kept at reset value.

Bit 2 **DV**: Data value comparator support

0: Single-shot data value comparisons not supported

Bit 1 **DA**: Data address comparator support

0: Single-shot data address comparisons not supported

Bit 0 **INST**: Instruction address comparator support

1: Single-shot instruction address comparisons supported

ETM single-shot processor comparator input control register (M7_ETM_SSPCIC0)

Address offset: 0x2C0

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PC[7:0]							
								r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PC[7:0]**: Comparator input selector for single-shot control

Selects one or more processor comparator inputs for single-shot control. One bit is provided for each processor comparator input.

ETM power-down control register (M7_ETM_PDC)

Address offset: 0x310

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PU	Res.	Res.	Res.
												r			

Bits 31:4 Reserved, must be kept at reset value.

Bit 3 **PU**: Power up request

Request to maintain power to the ETM and access to the trace registers.

0: Power not requested

1: Power requested

Bits 2:0 Reserved, must be kept at reset value.

ETM power-down status register (M7_ETM_PDS)

Address offset: 0x314

Reset value: 0x0000 0003

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	STICKY PD	POWER
														r	r

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **STICKYPD**: Sticky power-down state

This bit is set to 1 when power to the ETM registers is removed, to indicate that programming state has been lost. It is cleared after a read of the TRCPDSR.

0: Trace register power uninterrupted since the last read of PDS register

1: Trace register power interrupted since the last read of PDS register

Bit 0 **POWER**: ETM powered up

1: ETM is powered up; all registers are accessible

ETM claim tag set register (M7_ETM_CLAIMSET)

Address offset: 0xFA0

Reset value: 0x0000 000F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMSET[3:0]			
													rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value.

Bits 3:0 **CLAIMSET[3:0]**: Set claim tag bits

Write:

0000: No effect

xxx1: Set bit 0

xx1x: Set bit 1

x1xx: Set bit 2

1xxx: Set bit 3

Read:

0xF: Indicates there are four bits in claim tag

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMCLR[3:0]			
												rw	rw	rw	rw

1xxx: Clear bit 3

Read: Returns current value of claim tag

Reset value: N/A

[illegible]

Other values: Disable write access

ETM lock status register (M7_ETM_LSR)

Address offset: 0xFB4

Reset value: 0x0000 0003

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LOCKT YPE	LOCKG RANT	LOCKE XIST
													r	r	r

Bits 31:3 Reserved, must be kept at reset value.

Bit 2 **LOCKTYPE**: Size of the M7_ETM_LAR register

0: 32-bit

Bit 1 **LOCKGRANT**: Current status of lock

This bit always returns zero when read by an external debugger.

0: Write access is permitted

1: Write access is blocked. Only read access is permitted.

Bit 0 **LOCKEXIST**: Existence of lock control mechanism

The bit indicates whether a lock control mechanism exists. It always returns zero when read by an external debugger.

0: No lock control mechanism exists

1: Lock control mechanism is implemented

ETM authentication status register (M7_ETM_AUTHSTAT)

Address offset: 0xFB8

Reset value: 0x0000 000A

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SNID[1:0]		SID[1:0]		NSNID[1:0]		NSID[1:0]	
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:6 **SNID[1:0]**: Security level for secure non-invasive debug

0x0: Not implemented

Bits 5:4 **SID[1:0]**: Security level for secure invasive debug
0x0: Not implemented

Bits 3:2 **NSNID[1:0]**: Security level for non-secure non-invasive debug
0x2: Disabled
0x3: Enabled

Bits 1:0 **NSID[1:0]**: Security level for non-secure invasive debug
0x2: Disabled
0x3: Enabled

ETM CoreSight device architecture register (M7_ETM_DEVARCH)

Address offset: 0xFBC

Reset value: 0x4770 4A13

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ARCHITECT[10:0]											PRESEN	REVISION[3:0]			
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARCHID[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:21 **ARCHITECT[10:0]**: Component architect
0x23B: Arm®

Bit 20 **PRESENT**: Indicates the presence of this register
1: Present

Bits 19:16 **REVISION[3:0]**: Architecture revision
0x0: Rev 0

Bits 15:0 **ARCHID[15:0]**: Architecture ID
0x4A13: ETMv4 component

ETM CoreSight device type identity register (M7_ETM_DEVTYPE)

Address offset: 0xFCC

Reset value: 0x0000 0013

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SUBTYPE[3:0]				MAJORTYPE[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **SUBTYPE[3:0]**: Device sub-type identifier
0x1: Processor trace

Bits 3:0 **MAJORTYPE[3:0]**: Device main type identifier
0x3: Trace source

ETM CoreSight peripheral identity register 4 (M7_ETM_PIDR4)

Address offset: 0xFD0

Reset value: 0x0000 0004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]				JEP106CON[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **4KCOUNT[3:0]**: Register file size
0x0: Register file occupies a single 4 Kbyte region

Bits 3:0 **JEP106CON[3:0]**: JEP106 continuation code
0x4: Arm® JEDEC code

ETM CoreSight peripheral identity register 0 (M7_ETM_PIDR0)

Address offset: 0xFE0

Reset value: 0x0000 0075

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PARTNUM[7:0]**: Part number field, field, bits [7:0]
0x75: ETM part number

ETM CoreSight peripheral identity register 1 (M7_ETM_PIDR1)

Address offset: 0xFE4

Reset value: 0x0000 00B9

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **JEP106ID[3:0]**: JEP106 identity code field, bits [3:0]

0xB: Arm® JEDEC code

Bits 3:0 **PARTNUM[11:8]**: Part number field, bits [11:8]

0x9: ETM part number

ETM CoreSight peripheral identity register 2 (M7_ETM_PIDR2)

Address offset: 0xFE8

Reset value: 0x0000 001B

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC	JEP106ID[6:4]		
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVISION[3:0]**: Component revision number

0x1: r0p2

Bit 3 **JEDEC**: JEDEC assigned value

1: Designer ID specified by JEDEC

Bits 2:0 **JEP106ID[6:4]**: JEP106 identity code field, bits [6:4]

0x3: Arm® JEDEC code

ETM CoreSight peripheral identity register 3 (M7_ETM_PIDR3)

Address offset: 0xFEC

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVAND[3:0]**: Metal fix version

0x0: No metal fix

Bits 3:0 **CMOD[3:0]**: Customer modified

0x0: No customer modifications

ETM CoreSight component identity register 0 (M7_ETM_CIDR0)

Address offset: 0xFF0

Reset value: 0x0000 000D

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[7:0]**: Component ID field, bits [7:0]

0x0D: Common ID value

ETM CoreSight component identity register 1 (M7_ETM_CIDR1)

Address offset: 0xFF4

Reset value: 0x0000 0090

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **CLASS[3:0]**: Component ID field, bits [15:12] - component class
 0x9: Debug component with CoreSight-compatible registers

Bits 3:0 **PREAMBLE[11:8]**: Component ID field, bits [11:8]
 0x0: Common ID value

ETM CoreSight component identity register 2 (M7_ETM_CIDR2)

Address offset: 0xFF8

Reset value: 0x0000 0005

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[19:12]**: Component ID field, bits [23:16]
 0x05: Common ID value

ETM CoreSight component identity register 3 (M7_ETM_CIDR3)

Address offset: 0xFFC

Reset value: 0x0000 00B1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[27:20]**: Component ID field, bits [31:24]
 0xB1: Common ID value

Cortex-M7 ETM register map and reset values

The ETM registers are accessed by the debugger via the Cortex-M7 PPB, at address range 0xE0041000 to 0xE0041FFC.

Table 627. Cortex-M7 ETM register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0x004	M7_ETM_PRGCTL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	EN						
	Reset value																																0						
0x008	M7_ETM_PROCSSEL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PROCSEL[2:0]						
	Reset value																															0	0	0					
0x00C	M7_ETM_STAT	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PMSTABLE	IDLE					
	Reset value																															0	0	0					
0x010	M7_ETM_CONFIG	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DV	DA	Res	Res	Res	Res	RS	TS	COND[2:0]			VMD	CID	Res	CCI	BB	INSTP0[1:0]		Res					
	Reset value															0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	1					
0x014 to 0x01C	Reserved	Reserved																																					
0x020	M7_ETM_EVENTCTL0	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TYPE1	SEL1[3:0]				TYPE0		Res	Res	Res	SEL0[3:0]											
	Reset value																	0				0	0	0	0	0				0	0	0	0						
0x024	M7_ETM_EVENTCTL1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	LPOVERRIDE		ATB	Res	Res	Res	Res	Res	Res	INSTEN[3:0]									
	Reset value																				0		0							0	0	0	0						
0x02C	M7_ETM_STALLCTL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	DSTALL	ISATLL	Res	Res	Res	Res	Res	LEVEL[1:0]		Res	Res						
	Reset value																						0	0					0	0									
0x030	M7_ETM_TSCTL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TYPE	Res	Res	Res	Res	SEL[3:0]									
	Reset value																								0					0	0	0	0						
0x034	M7_ETM_SYNCP	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PERIOD[4:0]										
	Reset value																											0	1	0	1	0							
0x038	M7_ETM_CCCTL	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	THRESHOLD[11:0]																	
	Reset value																					0	0	0	0	0	0	0	0	0	0	0	0						
0x03C	Reserved	Reserved																																					
0x040	M7_ETM_TRACEID	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	TRACEID[6:0]													
	Reset value																										0	0	0	0	0	0	0						

Table 627. Cortex-M7 ETM register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x044 to 0x07C	Reserved	Reserved																																			
0x080	M7_ETM_VICTL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EXLEVEL_S3	Res.	Res.	EXLEVEL_S0	Res.	Res.	Res.	Res.	Res.	TRCERR	TRCRESET	SSSTATUS	Res.	TYPE	Res.	Res.	Res.	Res.	SEL[3:0]					
	Reset value													0			0						0	0	0		0				0	0	0	0			
0x084	Reserved	Reserved																																			
0x088	M7_ETM_VISSCTL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	STOP[7:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	START[7:0]											
	Reset value									0	0	0	0	0	0	0	0									0	0	0	0	0	0	0	0	0			
0x08C	M7_ETM_VIPCSSCTL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	STOP[3:0]				Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	START[3:0]						
	Reset value													0	0	0	0														0	0	0	0			
0x090 to 0x13C	Reserved	Reserved																																			
0x140	M7_ETM_CNTRLDV	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	VALUE[15:0]																				
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x144 to 0x17C	Reserved	Reserved																																			
0x180	M7_ETM_IDR8	MAXSPEC[31:0]																																			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0			
0x184	M7_ETM_IDR9	NUMP0KEY[31:0]																																			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x188	M7_ETM_IDR10	NUMP1KEY[31:0]																																			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x18C	M7_ETM_IDR11	NUMP1SPC[31:0]																																			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x190	M7_ETM_IDR12	NUMCONDKEY[31:0]																																			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1			
0x194	M7_ETM_IDR13	NUMCONDSPC[31:0]																																			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x198 to 0x1BC	Reserved	Reserved																																			
0x1C0	M7_ETM_IMSPEC0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SUPPORT[3:0]							
	Reset value																													0	0	0	0				
0x1E0	M7_ETM_IDR0	Res.	Res.	COMMOPT	TSSIZE[4:0]						Res.	Res.	Res.	Res.	Res.	Res.	QSUPP[1:0]				Res.	CONDTYPE[1:0]				NUMEVENT[1:0]				RETSTACK	TRCCCI	TRCCOND	TRCBB	Res.	Res.	Res.	Res.
	Reset value			0	0	1	1	0	0								0	0			0	1	1	1	1		1	1	1								

Table 627. Cortex-M7 ETM register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x1E4	M7_ETM_IDR1	DESIGNER[7:0]									Res.		Res.		Res.		Res.		Res.		Res.		Res.	TRCARCHMAJ[3:0]			TRCARCHMIN[3:0]			REVISION[3:0]			
	Reset value	0	1	0	0	0	0	0	1													0	1	0	0	0	0	0	0	0	0	0	1
0x1E8	M7_ETM_IDR2	Res.	Res.	Res.	CCSIZE[3:0]				DVSIZE[4:0]				DASIZE[4:0]				VMIDSIZE[4:0]				CIDSIZE[4:0]				IASIZE[4:0]								
	Reset value				0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0x1EC	M7_ETM_IDR3	NOOVERFLOW	NUMPROC[2:0]			SYSSTALL		STALLCTL	SYNCPR	TRCERR	Res.	Res.	Res.	Res.	EXLEVEL_S[3:0]			Res.	Res.	Res.	Res.	CCITMIN[11:0]											
	Reset value	0	0	0	0	0	1	0	1					1	0	0	1					0	0	0	0	0	0	0	0	0	1	0	0
0x1F0	M7_ETM_IDR4	NUMVMIDC[3:0]			NUMCIDC[3:0]			NUMSSCCC[3:0]			NUMRSPAIR[3:0]			NUMPC[3:0]			Res.	Res.	Res.	SUPPDAC		NUMDVC[3:0]			NUMACPAIRS[3:0]								
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0				0	0	0	0	0	0	0	0	0
0x1F4	M7_ETM_IDR5	REDFUNCNTR	NUMCNTR[2:0]		NUMSEQSTATE[2:0]			Res.	LPOVERRIDE	ATBTRIG	TRACEIDSIZE[5:0]					Res.	Res.	Res.	Res.	NUMEXTINSEL[2:0]		NUMEXTIN[8:0]											
	Reset value	1	0	0	1	0	0	0		1	1	0	0	0	1	1	1					0	1	0	0	0	0	0	0	0	0	1	0
0x1F8 to 0x204	Reserved	Reserved																															
0x208	M7_ETM_RSCTL2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PAIRINV	INV	Res.	GROUP[2:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SELECT[7:0]								
	Reset value										0	0		0	0	0									0	0	0	0	0	0	0	0	0
0x20C	M7_ETM_RSCTL3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	INV	Res.	GROUP[2:0]		Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SELECT[7:0]								
	Reset value										0		0	0	0										0	0	0	0	0	0	0	0	0
0x210 to 0x27C	Reserved	Reserved																															
0x280	M7_ETM_SSCC0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RST	Res.	Res.	PAIRINV	INV	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value								0																								
0x284 to 0x29C	Reserved	Reserved																															
0x2A0	M7_ETM_SSCS0	STATUS	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DV	DA	INST
	Reset value	0																												0	0	1	

Table 627. Cortex-M7 ETM register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0x2A4 to 0x2BC	Reserved	Reserved																																					
0x2C0	M7_ETM_SSPCIC0	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PC[7:0]													
	Reset value																										0	0	0	0	0	0	0	0					
0x2C4 to 0x30C	Reserved	Reserved																																					
0x310	M7_ETM_PDC	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PU	Res	Res						
	Reset value																														0								
0x314	M7_ETM_PDS	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	STICKYPD	Res						
	Reset value																															1	1						
0x318 to 0xF9C	Reserved	Reserved																																					
0xFA0	M7_ETM_CLAIMSET	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CLAIMSET[3:0]								
	Reset value																														1	1	1						
0xFA4	M7_ETM_CLAIMCLR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CLAIMCLR[3:0]								
	Reset value																														0	0	0						
0xFA8to 0xFAC	Reserved	Reserved																																					
0xFB0	M7_ETM_LAR	ACCESS_W[31:0]																																					
	Reset value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
0xFB4	M7_ETM_LSR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	LOCKTYPE	LOCKGRANT	LOCKEXIST						
	Reset value																														0	1	1						
0xFB8	M7_ETM_AUTHSTAT	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	SNID[1:0]			SID[1:0]			NSNID[1:0]			NSID[1:0]			
	Reset value																										0	0	0	0	1	0	1	0					
0xFBC	M7_ETM_DEVARCH	ARCHITECT[10:0]												PRESENT	REVISION[3:0]			ARCHID[15:0]																					
	Reset value	0	1	0	0	0	1	1	1	0	1	1	1	0	0	0	0	0	0	1	0	1	0	1	0	1	0	0	0	1	0	0	1	1					
0xFC0to 0xFC8	Reserved	Reserved																																					
0xFCC	M7_ETM_DEVTYPE	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	SUBTYPE[3:0]										MAJORITYTYPE[3:0]			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1					

Table 627. Cortex-M7 ETM register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0xFD0	M7_ETM_PIDR4	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	4KCOUNT[3:0]				JEP106CON[3:0]				
	Reset value																									0	0	0	0	0	1	0	0	
0xFD4 to 0xFDC	Reserved	Reserved																																
0xFE0	M7_ETM_PIDR0	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PARTNUM[7:0]								
	Reset value																									0	1	1	1	0	1	0	1	
0xFE4	M7_ETM_PIDR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	JEP106ID[3:0]				PARTNUM[11:8]				
	Reset value																									1	0	1	1	1	0	0	1	
0xFE8	M7_ETM_PIDR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	REVISION[3:0]				JEDC	JEP106ID[6:4]			
	Reset value																									0	0	0	1	1	0	1	1	
0xFEC	M7_ETM_PIDR3	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	REVAND[3:0]				CMOD[3:0]				
	Reset value																									0	0	0	0	0	0	0	0	
0xFF0	M7_ETM_CIDR0	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PREAMBLE[7:0]								
	Reset value																									0	0	0	0	1	1	0	1	
0xFF4	M7_ETM_CIDR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CLASS[3:0]				PREAMBLE[11:8]				
	Reset value																									1	0	0	1	0	0	0	0	
0xFF8	M7_ETM_CIDR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PREAMBLE[19:12]								
	Reset value																									0	0	0	0	0	1	0	1	
0xFFC	M7_ETM_CIDR3	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PREAMBLE[27:20]								
	Reset value																									1	0	1	1	0	0	0	1	

60.6.6 Cortex-M7 cross trigger interface (CTI)

See [Section 60.5.3](#).

60.7 References for debug infrastructure

1. IHI 0031C (ID080813) - Arm® Debug Interface Architecture Specification ADIv5.0 to ADIv5.2, Issue C
2. DDI 0480F (ID100313) - Arm® CoreSight™ SoC-400 r3p2 Technical Reference Manual, Issue G
3. DDI 0461B (ID010111) - Arm® CoreSight™ Trace Memory Controller r0p1 Technical Reference Manual, Issue B
4. DDI 0314H - Arm® CoreSight™ Components Technical Reference Manual, Issue H
5. DDI 0403D (ID100710) - Arm®v7-M Architecture Reference Manual, Issue E.b
6. DDI 0494-2a (ID062813) - Arm® CoreSight™ ETM™-M7 r0p1 Technical Reference Manual, Issue D